Tutorial 03

CS105.3 Database Management Systems

Answer All Questions

1. What is a database model?

2. What are the types of database models? Explain.

3. Identify the suitable database model to which the following scenarios belong.

Scenario 1:

A small local grocery store maintains a database to keep track of its inventory. In a single spreadsheet, each row represents a product, and columns contain information such as product name, quantity, and price. While simple, this approach becomes challenging as the store expands, making it difficult to manage relationships between suppliers, products, and sales.

Scenario 2:

An online retail company uses a database model to organize its product categories. The root node represents the general product categories (e.g., electronics, clothing), and each subsequent level branches into more specific subcategories. This structure helps in efficiently organizing and navigating through product information but might pose challenges when dealing with products that fall into multiple categories.

Scenario 3:

An airline reservation system employs a database model to manage flight schedules and passenger information. The database nodes represent flights, and edges define relationships between flights, connecting layovers and ensuring accurate passenger connections. This model allows for flexibility in representing complex relationships but requires careful management to avoid becoming overly intricate.

Scenario 4:

A small local event management company uses a database to store attendee information for each event. A single spreadsheet contains rows for each attendee, with columns for name, contact details, and event attended. As the company expands to organize multiple events simultaneously, the database structure becomes cumbersome, leading to difficulties in managing and analyzing data efficiently.

Scenario 5:

A university uses a database to store student information. Tables for students, courses, and grades are interlinked through key relationships. This enables the university to efficiently query and analyze data, ensuring that each student's grades are associated with the correct courses and professors. The model's structured approach facilitates data integrity and consistency.

Scenario 5:

A multimedia production company employs a database model to manage its digital assets. Each multimedia asset, such as images, videos, and audio files, is represented as an object with attributes like file type, resolution, and creation date. This approach allows for encapsulation of data and behavior within each asset, making it easier to maintain and extend the database as the company produces more diverse content.

Scenario 6:

An automobile manufacturing company employs a database model to organize its product assembly process. The top node represents the main product, such as a car model, and subsequent nodes represent components like engine, chassis, and electronics. While this structure simplifies tracking of the assembly process, changes in components can become challenging to implement without affecting the entire hierarchy.

Scenarios 7:

A large logistics company utilizes a database model to manage its global shipping operations. The database nodes represent various entities, including shipping routes, transportation hubs, and delivery points. Edges define the relationships between these entities, such as direct routes between hubs or specific delivery routes for different products. This model allows the company to efficiently track and optimize shipping routes, considering factors like cost, time, and capacity. However, the challenge lies in ensuring that changes in one part of the database do not adversely affect the entire logistics system, requiring careful management to maintain consistency and reliability in the data model.

Scenario 8:

A financial institution employs a database to manage customer accounts. Tables for customers, transactions, and account details are interconnected through relationships, ensuring that each transaction is accurately linked to the corresponding account. The model allows for efficient querying and reporting on financial data, ensuring accuracy and consistency in the management of customer accounts.